

Hassan Javaid Rao

AI Engineer · Senior Software Engineer — LLM Applications, Multi-Agent Systems & Voice AI

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PROFESSIONAL SUMMARY

AI-focused Senior Software Engineer with 5+ years of professional experience, including 4+ years building GCC enterprise logistics systems processing thousands of shipments daily across Oman, UAE, and Saudi Arabia. Specialized in **applied LLM engineering**: multi-agent pipelines, RAG systems, grounded AI assistants, and real-time AI voice agents. Initiated and lead the AI program at one of Oman's largest logistics platforms — shipping RAG chatbots, AI copilots, and a production multi-agent document-AI system that replaced most of a 12-hour daily manual data-entry workload — and independently built a **multi-tenant AI call-agent SaaS** that answers live phone calls with hallucination-resistant, knowledge-grounded responses. Strong foundation in scalable AWS architecture, multi-tenant SaaS, and technical mentorship. Section Leader at Stanford University's Code in Place program.

EXPERIENCE

Thiqa Tech — Muscat, Oman

Dec 2021 – Present

Promoted from Software Engineer to Senior Software Engineer; 5 years of continuous ownership of the same multi-tenant logistics SaaS platform.

► Senior Software Engineer

Jan 2024 – Present

- **AI initiative** Proposed, architected, and lead the platform's AI program — taking LLM features from concept to production for one of Oman's largest logistics companies (15 branches).
- Designed and shipped a **five-agent document-AI pipeline (LangGraph + Gemini)** that reads handwritten bilingual Arabic/English shipping labels into structured orders, cutting a 12-hour daily data-entry workload to 2–3 hours of review.
- Diagnosed the failure that killed the naive single-prompt version — **correct digits assigned to the wrong field** — and re-architected it as one agent per responsibility (region localization, OCR, location resolution, amounts, verification), making cross-field contamination structurally impossible.
- Built a **shadow-mode evaluation harness**: 2,700+ orders predicted alongside human operators and scored field-by-field, reaching **71.7% of orders billed identically to human entry**, with per-agent traces isolating whether an error came from reading or from resolution.
- Engineered cost and latency to **\$0.008 and 23s per order** through model routing (fast model reads, stronger model verifies) and deterministic pre-processing in Python; one targeted fix cut expensive verification escalations from 78% to 20% of orders.
- Built **RAG-based customer chatbots for shippers**: embedding + vector-retrieval pipelines over tenant shipment data and policy documents, with grounding rules that keep answers accurate, scoped, and hallucination-free.
- Shipped **"Ask AI" for operations** — a natural-language RAG assistant that answers any question about shipments, replacing minutes of manual cross-system lookups with instant grounded answers.
- Architected and led the **migration from a single monolith to a modular architecture** — defined module boundaries and drove incremental extraction with zero downtime.
- Engineered a high-performance export pipeline on **AWS Lambda + SQS** that reliably exports millions of records at minimal cost; built an internal real-time chat system used across branches (NestJS, MongoDB, Next.js).
- Mentor junior engineers through code reviews and architectural guidance; maintain 99.9% uptime with Sentry and CloudWatch observability.

► Software Engineer

Dec 2021 – Dec 2023

- Core engineer on a **multi-tenant logistics SaaS built from zero**, scaling it to thousands of shipments per day and millions of records across Oman, UAE, and Saudi Arabia.
- Designed the platform's **role-based access control (RBAC)** spanning tenants, branches, and operational roles.
- Built secure REST APIs with **OAuth 2.0** — Password Grant for first-party mobile apps, Client Credentials for machine-to-machine integrations with logistics partners **DHL, AJEX, and iMile**.
- Improved system performance ~20% by refactoring legacy code into modern patterns; introduced Redis-backed queues to keep UX smooth under heavy load.
- Stack: Laravel, Vue.js, Node.js, MySQL, Redis, AWS.

Freelance Full-Stack Developer — Fiverr & Upwork

2020 – 2021

- Top Rated on both platforms with **250+ five-star reviews**; delivered web and automation solutions for 200+ clients across logistics, e-commerce, and SaaS.

Section Leader (Teaching) — Stanford Code in Place

2025

- Competitively selected to teach Python to international learners in Stanford's flagship open-access program; led weekly sessions and gave personalized feedback.

PROJECTS

Waybill Data-Entry Agents — Multi-Agent Document AI (production)

Python (FastAPI), LangGraph, Google Gemini (Flash + Pro), Pillow, PostgreSQL + pgvector, Laravel, Docker

- Five single-responsibility agents on a **fixed LangGraph state graph** with parallel fan-out and a bounded verification/escalation loop — deliberately not a dynamic supervisor, so repeat runs of the same order stay comparable across versions.
- **Visual grounding before OCR**: a localizer agent returns box coordinates only, then Python handles rotation, cropping and upscaling — which also fixed a resolution failure where downsampling made handwriting unreadable before the model ever saw it.
- **Constrained selection over free-text generation** for location resolution: the agent picks an ID from a retrieved candidate list, making hallucinated destinations impossible on the field that drives delivery pricing.
- **RAG grounding** over similar past orders and prior customer records (pgvector), plus **LLM-as-judge verification** where a stronger model challenges each field and disagreements escalate to a wider crop; failure taxonomy separates provider errors from bad model output so outages mark entries retryable instead of storing empty predictions as successes.

AI Caller — Multi-tenant AI Voice-Agent SaaS (founder/solo build)

ai-caller.hassanrao.com

Python (FastAPI), Laravel, Vue.js, PostgreSQL + pgvector, Deepgram Voice Agent, Twilio, Google Gemini, AWS

- Built and launched a SaaS where any business self-serves an **AI phone receptionist** that answers real inbound calls 24/7 in a natural voice — live in production on AWS.
- Engineered the **real-time voice pipeline**: Twilio Media Streams bridged to Deepgram's agent API (STT → LLM → TTS) over WebSockets, with barge-in interruption handling and **LLM function calling** for shipment tracking, quote capture, and human transfer.
- Designed the **RAG grounding layer**: PDF knowledge bases chunked and embedded (Gemini) into pgvector with HNSW cosine indexes; tenant-scoped retrieval with relevance thresholds forces honest "I don't know" responses instead of hallucinations, with prompt-injection defenses around retrieved content.
- **LLM-automated onboarding**: when a company signs up in a new industry, Gemini researches the domain and writes that industry's agent rulebook (scope fences, compliance rules) automatically.
- Full SaaS product: self-serve signup with plans, per-tenant agent configuration, bring-your-own-Twilio with API-verified credentials and automatic webhook setup, and complete call transcripts with captured leads. Deployed on AWS EC2 (nginx, systemd) with GitHub Actions CI/CD.

Translation Management API

PHP 8.2, Laravel, MySQL, Redis, Docker

- Production-grade REST API for locale-based translations: Controller → Service → Repository architecture, Redis tag-based cache invalidation, and cursor-based export streaming millions of records at low memory cost; fully Dockerized with a PHPUnit suite covering API contracts, caching, and exports.

TECHNICAL SKILLS

AI / LLM Engineering	Multi-agent orchestration (LangGraph), RAG pipelines, embeddings & vector search (pgvector), vision & document AI, prompt engineering & grounding, constrained decoding, LLM function/tool calling, LLM-as-judge & offline evaluation, hallucination mitigation, voice agents (Deepgram, Twilio), Google Gemini APIs
Backend	Python (FastAPI), PHP (Laravel), Node.js (NestJS), TypeScript
Frontend	Vue.js, React.js / Next.js, Inertia.js, Tailwind CSS
Cloud & DevOps	AWS (Lambda, SQS, S3, EC2, CloudWatch), Docker, GitHub Actions CI/CD, nginx, Redis
Databases	PostgreSQL (+ pgvector), MySQL, MongoDB, Redis
Architecture	Multi-tenant SaaS, modular monolith & microservices, event-driven systems, REST API design, OAuth 2.0 / JWT
Quality & Monitoring	Sentry, AWS CloudWatch, PHPUnit, Git, Agile/Scrum

EDUCATION

B.S. Software Engineering — Sukkur IBA University

2017 – 2020

Fully funded merit-cum-need scholarship (4 years) • IELTS 7.0 • Coursework: AI, Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Web Engineering

AWARDS

Winner — Web Design Competition, Sukkur IBA • Winner — Code in the Dark, Sukkur IBA • 4-year fully funded merit scholarship